

- 17 A beam of light that consists of all wavelengths between 480 nm and 600 nm is projected on a diffraction grating that contains 500 lines per millimeter.

What is the maximum number of complete continuous spectra that can be observed emerging from the grating?

- A** 6 **B** 7 **C** 8 **D** 9

For longest wavelength: $n \lambda / d \leq 1 \rightarrow n \leq (1.0 \times 10^{-3} / 500) / 600 \times 10^{-9} = 3.3$

max $n = 3$

Thus there will be 6 spectra observed. (Zeroth order maximum is not a spectrum.)

- 18 An insulated rod with equal and opposite charges at its ends is placed in a non-uniform electric